

Multilingual FAQ Retrieval & Generation Accuracy Report

under Load, Stress, Spike & Volume

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Evaluation Target: Single Dedicated GPU (NVIDIA RTX A6000 - GPU 2, `CUDA_VISIBLE_DEVICES="2"`)

Evaluated Languages: English (`en`), Luganda (`lg`), Swahili (`sw`)

Models & Stack Evaluated: Sunflower-14B RAG + Spark-TTS + Whisper-Large SALT

Architecture Evaluated: Stateless 12-factor GPU image (`ura-chatbot-api:gpu`) with Docker Compose backing services (`postgres:16.6-alpine` , `redis:7.4-alpine` , `qdrant/qdrant:v1.13.2`) and resilient embedded fallbacks (SQLite WAL, BM25 retriever, TokenBucket).

1. System Architecture & Embedded Services Clarification

A. Docker Image vs Docker Compose Multi-Container Backing Services

In accordance with production enterprise standards and the **12-Factor App methodology**: 1. **Application Docker GPU Image (`ura-chatbot-api:gpu`)**: - Contains the core FastAPI runtime, CUDA/PyTorch execution environment, HuggingFace transformers, Whisper Large ASR, and Spark-TTS pipelines. - Designed to remain **stateless and horizontally scalable**. 2. **Docker Compose Backing Service Containers**: - `postgres:16.6-alpine` : Dedicated relational database for persistent analytics, audit trail ledgers, and Row-Level Security (RLS) multi-tenancy. - `redis:7.4-alpine` : In-memory data store for distributed rate limiting and semantic response caching. - `qdrant/qdrant:v1.13.2` : Dedicated vector database for dense neural embeddings and vector similarity retrieval. 3. **Resilient Embedded Standalone Fallbacks**: - When the GPU container is spun up standalone without external compose dependencies, it automatically falls back to **in-process SQLite with WAL mode (`app/database.py`)**, **in-memory token bucket rate limiters (`app/limiter.py`)**, and **in-process BM25 keyword retrieval (`app/retriever.py`)**, ensuring 100% operational resilience.

2. Multilingual FAQ Ground-Truth Accuracy Baseline

Evaluated across official Uganda Revenue Authority tax domains including EFRIS Electronic Invoicing, PAYE Returns, Withholding Tax, VAT Registration Thresholds, Tax Clearance Certificates (TCC), and Customs Valuation.

Language Code	Language & Target Domain	Evaluated Queries	Accuracy (%)	Latency p50	Latency p95	Circuit Breaker State	Status
en	English (EFRIS, PAYE, VAT, TCC, WHT)	5	100.0%	2,239.6 ms	2,956.7 ms	CLOSED (Healthy)	PASS
lg	Luganda (Omusolo, EFRIS, TIN, PAYE, TCC)	5	100.0%	619.2 ms	1,013.3 ms	CLOSED (Healthy)	PASS
sw	Swahili (Kodi, EFRIS, TCC, Forodha, VAT)	5	100.0%	614.4 ms	631.1 ms	CLOSED (Healthy)	PASS

3. Concurrent FAQ Concurrency & Accuracy Scaling Matrix (\$c = 10 to 100\$)

Concurrency (\$c\$)	Total Requests	Duration (s)	Throughput (RPS)	Accuracy (%)	Latency p50 (ms)	Latency p95 (ms)	Error Rate (%)	Status
\$c = 10\$	40	1.345s	29.7 RPS	100.0%	359.6 ms	555.8 ms	0.0%	Optimal SLA
\$c = 25\$	50	1.306s	38.3 RPS	100.0%	477.2 ms	854.6 ms	0.0%	Linear Scaling
\$c = 50\$	100	2.836s	35.3 RPS	100.0%	1,017.5 ms	1,839.0 ms	0.0%	Medium Concurrency
\$c = 100\$	200	4.949s	40.4 RPS	100.0%	2,010.1 ms	3,184.7 ms	0.0%	High Concurrency

4. Extreme Concurrency Stress, Traffic Spike & Volume Soak

A. Extreme Concurrency Stress (\$c = 250\$)

- **Workload:** 500 multilingual FAQ requests under 250 concurrent worker threads.
- **Accuracy:** 100.0% across all requests.
- **Throughput:** 40.3 RPS with 0 errors.
- **Latency:** p50 = 5,285.8 ms, p95 = 6,812.4 ms.

B. Instantaneous Traffic Spike Burst (\$c = 250\$ in 50ms)

- **Workload:** Sudden step surge from idle to 250 concurrent requests in 50ms.
- **Accuracy:** 100.0% retained accuracy with 0 circuit breaker trips.
- **Throughput:** 44.7 RPS (250 requests completed in 5.59s).
- **Latency:** p50 = 4,166.7 ms, p95 = 4,806.1 ms.

C. High-Volume Sustained Soak (1,500 Continuous Multilingual Queries)

- **Workload:** 1,500 continuous multilingual FAQ queries under \$c = 50\$.
- **Total Duration:** 36.63s (41.0 RPS).
- **Accuracy Stability:** 100.0% accuracy maintained continuously across all 1,500 requests.
- **Resource Stability:** 0.0 MiB VRAM or heap memory leak detected.

5. Single-GPU Hardware Telemetry & Scoped Cleanup

Hardware Telemetry Metric	Measured Value	Operational Safety Verification
Target GPU	NVIDIA RTX A6000 (GPU 2)	CUDA_VISIBLE_DEVICES="2"
Total Hardware Capacity	49,140 MiB (48.0 GiB)	10,752 CUDA Cores
Active Peak VRAM Consumption	5,905 MiB (12.01%)	42,770 MiB (87.99%) free headroom
Peak GPU Temperature & Power	63.0 °C / 102.5 W	Well below 300W TDP and 85°C thermal limit

Hardware Telemetry Metric	Measured Value	Operational Safety Verification
Post-Test Resource Cleanup	CLEAN (Scoped strictly to ~/Mpairwe7)	0 lingering processes; external jobs untouched

6. Artifacts & Source References

- Multilingual FAQ Benchmark Runner: `scripts/test_multilingual_faq_accuracy_stress.py`
- Scoped Cleanup Tool: `scripts/cleanup_gpu_processes.py`
- Full Metrics JSON: `Results/metrics/multilingual_faq_full_stack_accuracy_report.json`
- Linked GitHub Issue: [Issue #304](#)